

Recording a route without satellites

How accurate a phone’s own sensors are, measured against GPS

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Abstract

When GPS is lost — in a tunnel, an underground car park, or an aircraft — a phone can still draw the route it travels. It reads *speed* from how the vehicle shakes and *direction* from its gyroscope and compass, and it never uses GPS after the first point. This paper measures the newest version of that system on a motorcycle, in a car, on foot and on a plane, by running it over every earlier recording and comparing the result with GPS recorded at the same time, used only as the answer key. Direction works in all four: it was within 30° of GPS 72% of the time on a motorcycle, 69% in a car, 93% on foot and 99% on a flight. Over 72 journeys (19 hours), distance came out 10% short on a motorcycle and 1% short in a car. Walking came out 4% long, and in the air speed cannot be measured at all. On a motorcycle, fast riding reads slow and slow riding reads fast. A phone with few examples of its own uses a small built-in network instead; on recordings it had never seen, it was as accurate as a phone with a full store.

1 What the system does

A route is a series of short steps. Each step needs a length and a direction. With GPS both come from satellites; here both come from the phone’s motion sensors.

Speed from vibration. A moving vehicle shakes, and it shakes differently at different speeds [1]. (Adding up acceleration instead does not work on a phone: its small sensor errors grow into kilometres within minutes [2].) Every second, the phone takes the last five seconds of up-and-down acceleration, measured 50 times a second, and turns it into a short “fingerprint” of eleven numbers — how much energy the shaking has at each of nine frequency ranges, plus two measures of its overall strength. Whenever GPS is available, the phone stores fingerprints together with the speed GPS measured at that moment; it keeps up to 4000 of them. Without GPS, it looks up the 12 stored fingerprints closest to the current one and averages their speeds:

$$\hat{v} = \frac{\sum_{j=1}^{12} v_j / d_j^2}{\sum_{j=1}^{12} 1 / d_j^2} \quad (1)$$

where v_j is a stored speed and d_j how far its fingerprint is from the current one. If even the closest stored fingerprint is far away, the phone does not guess. Averaging tends to pull every answer toward the middle, so a correction learned from the stored examples stretches the answers back out.

A starting model. The stored examples come from the phone’s own trips, so a new phone, or one whose store has been cleared, has none. Until it has 3000, it uses a small network instead: the same eleven-number fingerprint goes in, a speed comes out. The network was trained on recorded trips and ships with the app, so it needs nothing from the current trip. Like the store, it declines a fingerprint unlike anything it was trained on.

Direction from turns. The gyroscope measures every turn precisely but does not know north; the compass knows north but measures where the *phone* points, not where the vehicle is going — a known problem for phones carried in different ways [3]. In a pocket those differ by some angle β , which depends on how the phone happens to sit. The phone learns β from the vehicle’s own turns. In a turn the vehicle is pushed toward the inside of the curve, so the sideways push the phone feels, compared with which way the gyroscope says it is turning, shows where “forward” is. Written as complex numbers,

$$\hat{\beta} = \arg \sum_t \frac{a_t}{|a_t|} \overline{i v_t \omega_t} \quad (2)$$

where a_t is the sideways and forward acceleration measured relative to the compass, v_t the speed and ω_t the turn rate. The phone makes a first estimate after 30 seconds of riding and keeps improving it; when it improves, the part of the ride already drawn is redrawn with the better value.

Walking is handled differently: over four seconds, the back-and-forth of the steps lines up with the direction of walking, and each push-off is sharper than each landing, which says which end of that line is forward.

The compass reading itself. The phone’s built-in heading is unreliable in one common pocket position: when the phone sits head-down at an angle, it scattered by about 49° a minute even though the phone was not moving in the pocket. So the system computes its own heading from the phone’s orientation, using whichever edge of the phone is closest to level. That heading scattered by about 1° a minute in the same position.

Only the first point uses GPS. After that, each step is laid down from the previous point, along the current direction, for the distance the speed covers in that second.

2 How it was tested

GPS was recorded alongside every test and never used by the system; it serves only as the answer key. So that every result describes the newest version, the newest method was run again over all earlier recordings, which were made with the phone’s raw sensor readings saved — its acceleration, rotation and orientation angles, fifty times a second:

- **Speed.** The app’s speed logic was run again on the recorded acceleration: the same fingerprint, the same 12-neighbour average, the same refusals and correction, and the same rules for a stopped vehicle, for a phone being handled, and for holding the last answer. Its memory holds 4000 examples drawn at random from every *other* journey,

so that no journey was ever measured with its own GPS in the memory. On ten recent journeys the re-run came out typically 5% shorter than what the app itself had drawn.

- **Direction.** The direction method, including the redrawing, was run again on the recorded rotation and orientation, with the re-run speed.
- **Starting network.** Each recording was measured by a network trained without it (the recordings were split into eight groups), and the store was tested holding only 60 to 4000 examples.
- **Walking.** The app's step detector was run again on the recorded acceleration, with the stride the phone had learned. The phone's own step counter, which the app also consults, was not recorded; its share is taken from each journey's first walk, where it added about 2% on average.

Three inputs are taken from the recordings as they were: whether each second was walking or riding (which older versions decided partly with GPS help the current one no longer uses), Apple's motion classifier, and the car-park ramp flag. Cabin pressure was not recorded, so on the flight the app's switch to examples learned on earlier flights could not be re-run.

Every result is split by what was moving. A motorcycle and a car were told apart from the sensors themselves: in a turn a motorcycle leans and the phone leans with it, while a car stays level and the phone is pushed sideways. This agreed with every journey whose vehicle was known. Walking is the time the app was counting steps. That gives 39 motorcycle journeys (11 hours, the phone in a trouser pocket) and 33 car journeys (7.5 hours, the phone in a pocket or lying flat) for speed and distance, 66 routes, 64 recordings for direction, 43 straight half-minutes and seven longer walks on foot, and one flight. Half-minutes that mix walking and riding, or in which the app counted steps while GPS showed riding speed, are left out.

In a pocket GPS is itself only accurate to 10–40 m, so it was used over stretches long enough for that not to matter:

- **Speed and distance** are compared over half-minute stretches between GPS fixes that are better than 40 m: the straight distance GPS moved, against the distance the system added. Walking moves so little in half a minute that only straight stretches between fixes better than 25 m are used.
- **Direction** is compared every second: the bearing from the GPS position 15 s earlier to the one 15 s later, against the system's heading over the same half-minute. For the plane, which covers far more ground, the stretch is ten seconds.
- **Whole routes** are compared by finding the rotation and the size change that best lay the drawn route over the GPS track, pinned at the shared starting point. The rotation is how far the whole route is turned; the size change is how much too big or too small it is.

3 Results

Every graph is split by what was moving: a motorcycle, a car, a person walking, or a plane.

3.1 Speed

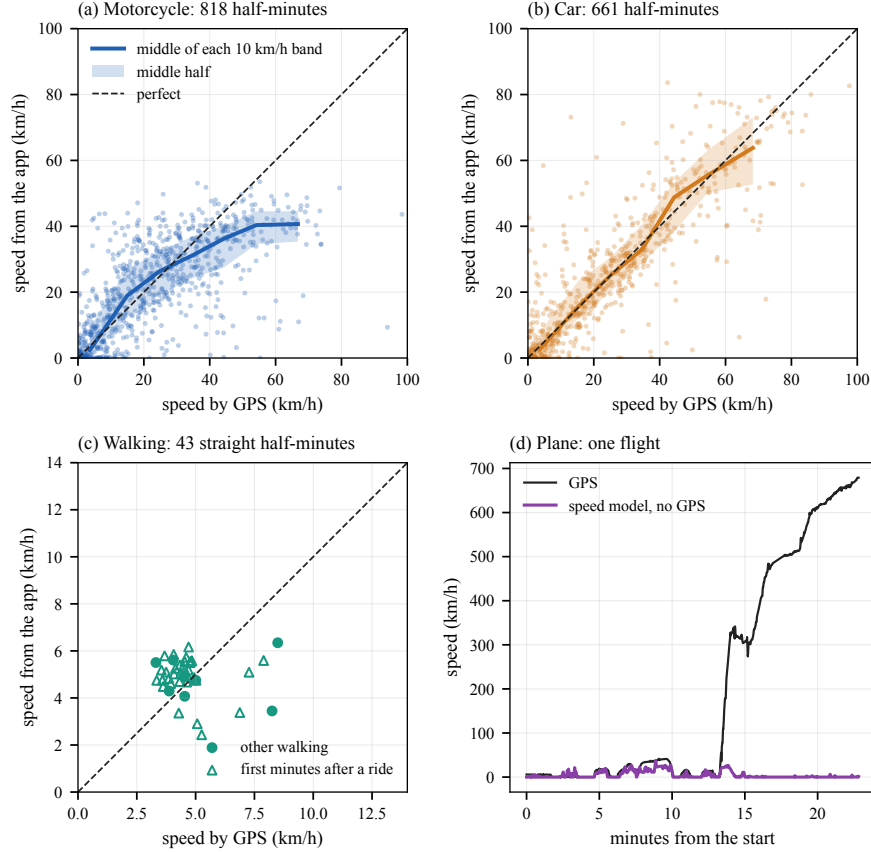


Figure 1: Speed from the app against speed from GPS. (a, b) Each dot is a half-minute of riding; the solid line is the middle of each 10 km/h band and the shading holds the middle half. On the dashed line the two agree. (c) Straight half-minutes of walking; triangles are the first minutes after a ride. (d) One flight, over time.

- **Motorcycle** (818 half-minutes). The app squeezes the range: 10–20 km/h reads about 19, 20–40 km/h is close, and above 60 km/h it reads about 41 where GPS measured 66. The squeeze comes from averaging twelve examples: when a fingerprint could belong to several speeds, the average lands in the middle. The average error is 9 km/h.
- **Car** (661 half-minutes). Much closer: from 10 to 80 km/h the middle reading is within about 5 km/h of GPS. Only above 80 km/h does it read low (71 where GPS measured 92). The average error is 7.5 km/h.

- **Walking** (43 straight half-minutes). About 5 km/h where GPS measured about 4.5, just after a ride as at any other time.
- **Plane** (one flight). While taxiing the app reads low (about 12 km/h where GPS measured 18). In the air the cabin is smooth, and the speed model reads it as almost standing still.

3.2 Distance

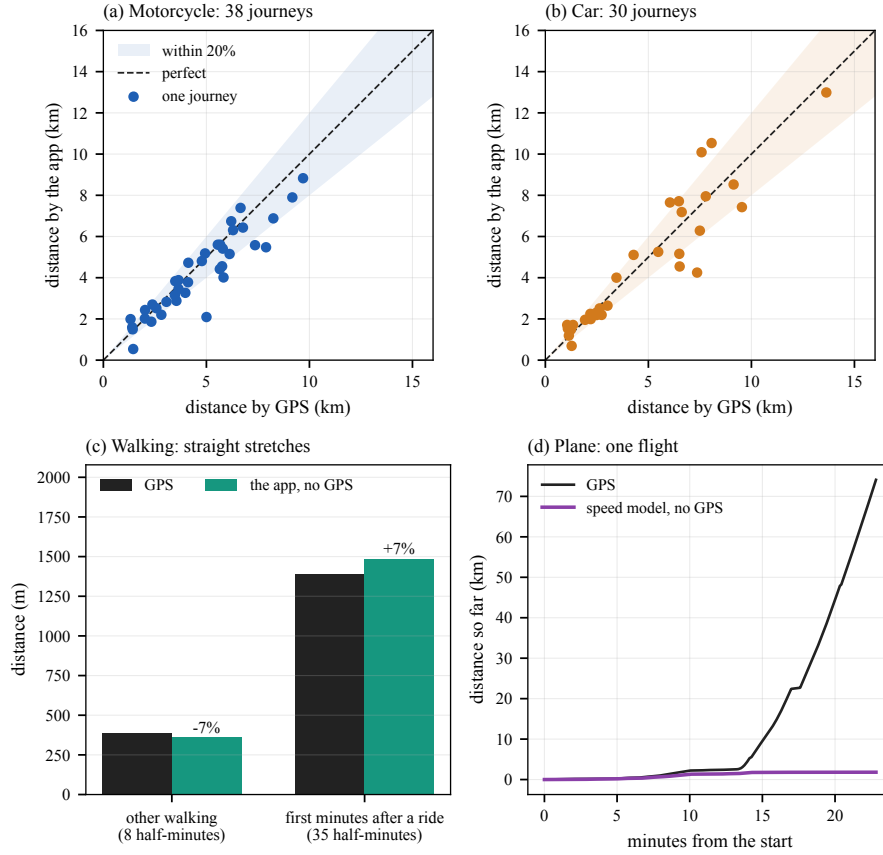


Figure 2: Distance by the app against GPS. (a, b) Each dot is one journey with at least 1 km that GPS could check; the shading is within 20%. (c) Straight stretches of walking, added up. (d) Distance so far on one flight.

- **Motorcycle** (39 journeys, 177 km checked). The app added 160 km, 10% short: the distance lost at high speed outweighs what is gained at low speed. Of the 38 journeys with at least 1 km checked, the middle one was 6% short and 74% were within 20%; the worst were 63% short and 51% long.
- **Car** (33 journeys, 145 km checked). The app added 144 km, 1% short. Of the 30 journeys with at least 1 km checked, the middle one was 1% short and 63% were within 20%; the worst were 46% short and 61% long.

- **Walking.** Over straight stretches the app was 4% long: 7% long in the first minutes after a ride and 7% short at other times.
- **Plane.** The speed model counted 1.8 km of the 74 km flown.

Motorcycle and car together, the app added 304 km where GPS measured 322 km, 5.5% short.

3.3 Direction

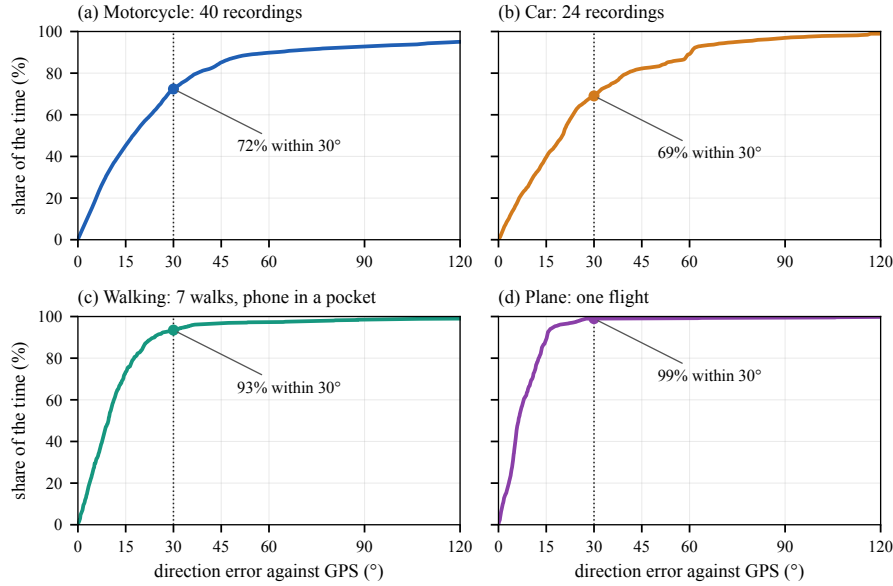


Figure 3: How often the direction is within a given angle of GPS, compared every second with a half-minute of GPS track (ten seconds for the plane).

Direction holds up in every case (Figure 3):

- **Motorcycle** (40 recordings): within 30° 72% of the time; middle error 17°.
- **Car** (24 recordings): within 30° 69% of the time; middle error 20°.
- **Walking** (seven walks with the phone in a pocket): within 30° 93% of the time; middle error 10°.
- **Plane** (one flight): within 30° 99% of the time; middle error 6°.

3.4 Whole routes

Figure 4 and Table 1 put speed and direction together.

- **Motorcycle** (40 routes, 349 km, many with a walk at one end). The median route is turned 8° and 85% are within 30°. Routes come out small — a median 0.85 of the real size — partly because distance is short, and partly because small direction errors

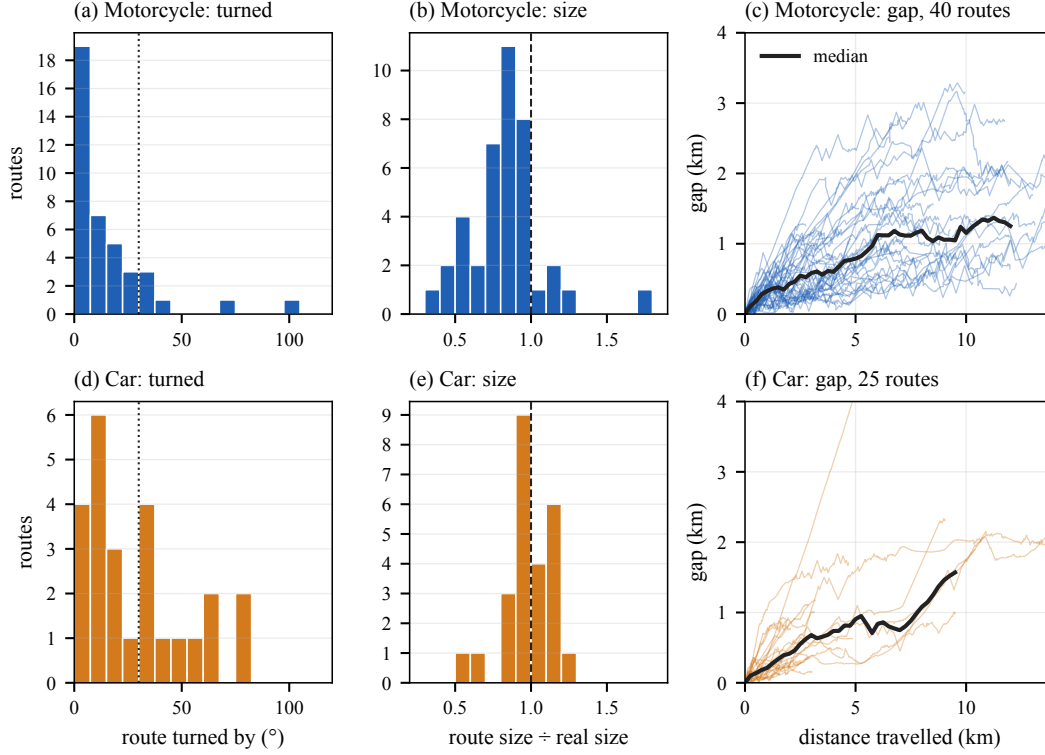


Figure 4: Routes drawn by the app, compared with GPS. (a, d) How far each whole route is turned from the GPS track; the dotted line is 30° . (b, e) Its size relative to the GPS track. (c, f) How far the drawn position is from the GPS position as the journey goes on: one thin line per route, and the median.

Table 1: Summary: the newest version against GPS.

	Distance	Direction within 30°	Whole route turned
Motorcycle	10% short; 74% of journeys within 20%	72% of the time	median 8°
Car	1% short; 63% of journeys within 20%	69% of the time	median 20°
Walking	4% long	93% of the time	—
Plane	1.8 of 74 km	99% of the time	—

make the path fold. The drawn position drifts from GPS by a median of 17% of the distance travelled.

- **Car** (25 routes, 126 km). The median route is turned 20° and 56% are within 30° ; the size is right (median 0.98). The drift is 18% of the distance travelled.
- **Walking**. Only one route was mostly walking, 2.5 km long; it was turned 53° .
- **Plane**. No route can be drawn, because speed in the air is not measured.

Nine routes are turned more than 45° . Seven of them are short journeys, under 3 km, with

too few turns to learn the pocket angle well.

3.5 The flight, second by second

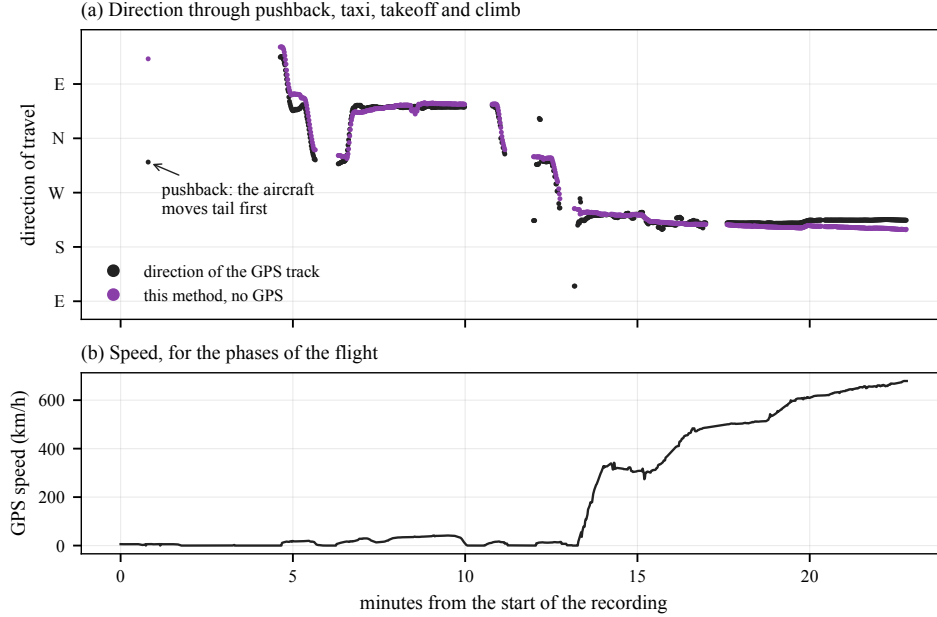


Figure 5: One flight, from the gate to the climb. (a) The direction of travel from the GPS track (black) and from this method, which uses no GPS (purple). (b) The aircraft’s speed from GPS, to show where it was taxiing, taking off and climbing.

Figure 5 follows the flight with the same direction method as on the road. While taxiing, the middle error was 6° ; in the air it was 7° . Nine seconds were more than 30° off. One is the pushback at the start, when the aircraft moves tail first: the method gives the way the nose points, which is opposite to the way it moves. The other eight came while the aircraft was stopped or crawling before the runway, where the GPS track has no clear direction. In the climb the gap grows slowly, to about 13° by the end of the recording. Part of that may be wind, which pushes an aircraft sideways so that its nose and its track differ; GPS alone cannot tell the two apart.

3.6 A new phone

A phone with few examples of its own reads speed poorly: with 60 stored examples the average error was 10.6 km/h and only half of the journeys came within 20%. The built-in network, measured only on recordings it had never seen, averaged 8.3 km/h, and 75% of journeys were within 20% — as good as a full store of 4000 examples (8.4 km/h, 69%). The store draws level at about 3000 examples, so that is where the app hands over (Figure 6). On a motorcycle the network averaged 9.1 km/h and in a car 7.2 km/h, against 9.1 and 7.5 for a full store.

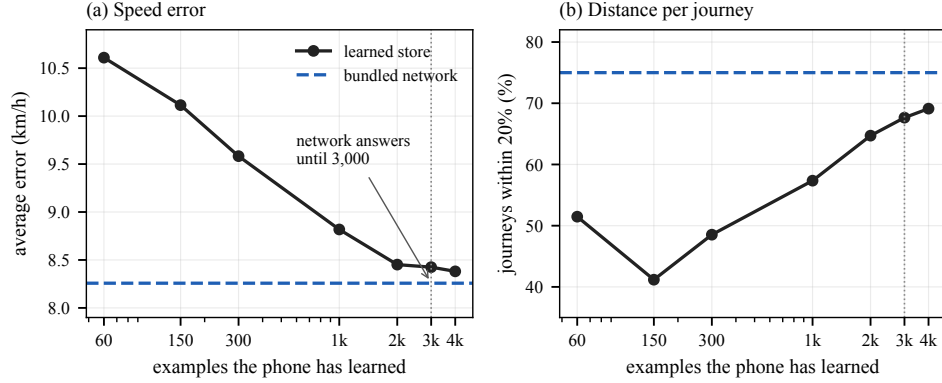


Figure 6: A phone that has learned only a few examples, against the built-in network. The store was tested holding 60 to 4000 examples; the network was measured only on recordings it had never seen. The dotted line is where the app hands over from the network to the store.

4 Where it goes wrong

- **On a motorcycle, fast riding reads slow and slow riding reads fast.** Above 60 km/h the speed is under two-thirds of the true value (Figure 1a). Over a whole journey the two partly cancel, but a journey that is mostly fast comes out short. A car does not have this problem below 80 km/h.
- **The first minute of a ride.** Until the pocket angle is learned, the direction comes from the compass alone, which in a pocket can be far off. The redrawing fixes the saved route afterwards, but a very short ride may never learn the angle well.
- **A phone that moves.** The pocket angle is assumed to stay the same for the whole ride. If the phone shifts, the rest of the ride is turned by about as much as it moved. When the phone is taken out of the pocket, the hand's movement hides the vehicle's vibration, so the last speed is held; on one ride, 48 km/h was held for ten seconds after the motorcycle had stopped, about 130 m that was never travelled.
- **Car-park ramps.** A car crawling up a concrete ramp shakes like one driving at 40 km/h, so distance is not counted while climbing and turning together. The same test can fire on an ordinary rising, curving road: once for nine seconds at 48 km/h, about 115 m lost.
- **Aircraft.** Speed cannot be measured in an aircraft. In the air the cabin is smooth, so a speed model that has learned only from roads reads it as nearly standing still (Figure 1d): 1.8 of 74 km. The app watches cabin pressure and, once it sees a climb, stops using its road examples and uses only examples learned on earlier flights that had GPS; with one flight recorded, that could not be tested. The phone's own orientation filter also treats the long, steady push of a takeoff as a tilt [4]. The direction still works (Figure 5).

5 What did not work

Many ideas were built and tested against GPS before the current version. The most instructive:

- **Neural networks.** A network that reads 20–40 seconds of fingerprints at once was more accurate on test data (8.8 against 10.6 km/h) and ran in the app for a while, but after it was retrained it under-read a pocketed ride by a third, and it was removed. The small network now used to start a new phone (Section 3.6) is different: it is never retrained on the phone, reads one fingerprint at a time, declines unfamiliar ones, and hands over to the store once the phone has learned enough of its own.
- **Direction from speeding up and braking.** The push when accelerating in a straight line points forward, so it should give direction. It was 44° off; turns work far better.
- **Other direction ideas:** the axis the motorcycle leans about (25° off), the direction of the rider’s thigh (73° off), and the swing of the thigh for telling forward from backward when walking (wrong more often than right).
- **How examples were forgotten.** When the store was full, it used to throw away examples from whichever speed it held most of, to keep rare fast examples. That left it holding 5% of its examples below 5 km/h when 24% of real riding is that slow, so slow riding read far too fast. Throwing away a random example instead fixed most of it.

6 Lessons

- **The phone’s own definitions caused more error than the models.** Its built-in heading becomes unreliable in a common pocket position, and its “user acceleration” points the opposite way to the vehicle’s acceleration. Each of these, once found, mattered more than any change to the estimators.
- **A test must not use what it is testing without.** Twice during development the no-GPS test mode quietly used GPS for direction, which made its results look better than they were.
- **Check what a counter stores while nothing reads it.** The step detector kept counting while the phone rode in a pocket — about one bump a second — and only a walk ever emptied it. The next walk then paid out hundreds of metres at twice the walking pace. Splitting the results by kind of travel is what exposed it: walks right after a ride came out twice their length. A ride now clears the count.
- **Check what a learning system remembers.** The store’s way of forgetting examples was sensible on its own and still shifted every answer, and it went unnoticed until its contents were compared with the riding they came from.

Limits of this evaluation

All recordings were made by one person, on one iPhone, mostly in one city, by motorcycle, car, on foot and on only one flight; other people, phones and vehicles may behave differently. That applies most to the starting network, which was tested on the same person’s other recordings: on someone else’s phone and vehicle it may do worse. Most results come from running the newest method again over earlier recordings rather than from live use, though on the journeys the app itself recorded, the re-run came out typically 5% shorter than the app had drawn. GPS in a pocket is itself only accurate to tens of metres, so small errors cannot be measured reliably.

References

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- [2] V. Capuano, L. Xu and J. Estrada Benavides. Smartphone MEMS accelerometer and gyroscope measurement errors: laboratory testing and analysis of the effects on positioning performance. *Sensors*, 23(17):7609, 2023.
- [3] W. Li, R. Chen, Y. Yu, Y. Wu and H. Zhou. Pedestrian dead reckoning with novel heading estimation under magnetic interference and multiple smartphone postures. *Measurement*, 182:109610, 2021.
- [4] M. B. Del Rosario, N. H. Lovell and S. J. Redmond. Quaternion-based complementary filter for attitude determination of a smartphone. *IEEE Sensors Journal*, 16(15):6008–6017, 2016.